

Problem 1.

- (i) What does it mean to say that a subset $H \subset G$ of a group is a *subgroup*?

Solution. We say H is a subgroup if:

- (S1) $e \in H$;
- (S2) $h \in H \Rightarrow h^{-1} \in H$;
- (S3) $h, h' \in H \Rightarrow h \cdot h' \in H$.

- (ii) Let $\alpha : X \rightarrow X$ be a function from a set to itself. Show that¹

$$\text{Cent}(\alpha) = \{\sigma \in \text{Sym}(X) : \sigma \circ \alpha = \alpha \circ \sigma\}$$

is a subgroup of $\text{Sym}(X)$.

Solution. Here is a direct verification:

- We have $\text{id}_X \circ \alpha = \alpha = \alpha \circ \text{id}_X$ so $\text{id}_X \in \text{Cent}(\alpha)$.
- If $\sigma \in \text{Cent}(\alpha)$, then $\sigma^{-1} \in \text{Cent}(\alpha)$ because

$$\sigma^{-1} \circ \alpha = \sigma^{-1} \circ \alpha \circ \sigma \circ \sigma^{-1} = \sigma^{-1} \circ \sigma \circ \alpha \circ \sigma^{-1} = \alpha \circ \sigma^{-1}.$$

- If $\sigma, \sigma' \in \text{Cent}(\alpha)$, then $\sigma \circ \sigma' \in \text{Cent}(\alpha)$ because

$$\sigma \circ \sigma' \circ \alpha = \sigma \circ \alpha \circ \sigma' = \alpha \circ \sigma \circ \sigma'.$$

¹Recall that $\text{Sym}(X)$ denotes the set of bijective functions $X \rightarrow X$, regarded as a group under the binary operation of composition.

Problem 2. Let G and G' be groups.

- (i) What does it mean to say that a function $\phi : G \rightarrow G'$ is a *group homomorphism*?

Solution. We say ϕ is a homomorphism if:

- (H1) $\phi(e_G) = e_{G'}$;
- (H2) $\phi(g^{-1}) = \phi(g)^{-1}$ for all $g \in G$;
- (H3) $\phi(g \cdot g') = \phi(g) \cdot \phi(g')$ for all $g, g' \in G$.

- (ii) Suppose now that $\phi : G \rightarrow G'$ is a group homomorphism, and that $H' \subset G'$ is a subgroup. Show that $\phi^{-1}(H')$ is a subgroup of G .

Solution. Here is a direct verification:

- We have

$$\phi(e_G) = e_{G'} \in H'$$

by (H1) and (S1) so $e_G \in \phi^{-1}(H')$.

- Let $h \in \phi^{-1}(H')$, i.e. $\phi(h) \in H'$. Then

$$\phi(h^{-1}) = \phi(h)^{-1} \in H'$$

by (H2) and (S2), so $h^{-1} \in \phi^{-1}(H')$.

- Let $h_1, h_2 \in \phi^{-1}(H')$, i.e. $\phi(h_1), \phi(h_2) \in H'$. Then

$$\phi(h_1 \cdot h_2) = \phi(h_1) \cdot \phi(h_2) \in H'$$

by (H3) and (S3), so $h_1 \cdot h_2 \in \phi^{-1}(H')$.

(iii) With notation as in (ii), show for every sequence of elements $g_1, \dots, g_r \in G$ that²

$$\phi(\langle g_1, \dots, g_r \rangle) \subset H' \quad \Leftrightarrow \quad \phi(g_1), \dots, \phi(g_r) \in H'.$$

Solution. This holds because:

$$\begin{aligned} \phi(\langle g_1, \dots, g_r \rangle) \subset H' & \\ \Leftrightarrow \langle g_1, \dots, g_r \rangle \subset \phi^{-1}(H') & \quad (\text{HW0}) \\ \Leftrightarrow g_1, \dots, g_r \in \phi^{-1}(H') & \quad (\text{by } (*) \text{ for the subgroup } \phi^{-1}(H') \subset G) \\ \Leftrightarrow \phi(g_1), \dots, \phi(g_r) \in H' & \quad (\text{definition}) \end{aligned}$$

²Recall that when $k_1, \dots, k_r \in K$ are elements of a group, $\langle k_1, \dots, k_r \rangle$ is the subgroup uniquely characterized by the property: for every subgroup $K' \subset K$, one has

$$\langle k_1, \dots, k_r \rangle \subset K' \quad \Leftrightarrow \quad k_1, \dots, k_r \in K'. \quad (*)$$

Problem 3. Find an integer x satisfying

$$[x]_{15} = [8]_{15} \quad \text{and} \quad [x]_{29} = [24]_{29}.$$

Solution. We compute

$$2 \cdot 15 - 29 = 1$$

so one solution is

$$x = -7 \cdot (-29) - 5 \cdot (2 \cdot 15) = 53.$$